Cyclistic bike-share analysis case study

This is my deliverable for the capstone project for Google Data Analytics Professional Certificate program.

Scenario
I am a junior analyst working for Cyclistic, a bike-share company in Chicago, as a member of their marketing analyst team. They believe that maximizing the number of annual memberships (converting casual memberships to annual memberships) will be the key to their future success. Our team aims to understand how casual riders and annual members utilize Cyclistic bikes differently.

With the insights gained from analyzing the data, we will design a new marketing strategy to convert casual riders into annual members. My recommendations must be approved by Cyclistic executives first

I will answer their business questions through the following six steps of the data analysis process: Ask, Prepare, Process, Analyze, Share, and Act.

1. Ask

The following three questions need to be answered to guide the future marketing program:

1) How do annual members and casual riders use Cyclistic bikes differently?
2) Why would casual riders buy Cyclistic annual memberships?
3) How can Cyclistic use digital media to influence casual riders to become members?

My understanding of the business task is to find the difference in the use of Cyclictic bikes between casual riders and annual members, analyze the data, and make recommendations based on the insights coming from the analysis of the data to be shared with the marketing team and Cyclistic executives.

2. Prepare

I downloaded the monthly data of Cyclistic trip data from Jan 2025 to Dec 2025 to .
The data,”divvy-tripdata” is located at https://divvy-tripdata.s3.amazonaws.com/index.html.
The data is stored in csv format and it has the following columns of data:

Column	Meaning
ride_id	rider id
rideable_type	types of bike, including classic bike and electric bike.
started_at	starting time of the bike use
ended_at	ending time of the bike use
ride_length (insterted)	length of each ride calculated by subtracting the data for the column of “started_at” from the column of “ended_at”
day_of_week(inserted)	Day of the week returned by the Excel weekday function from 1

	(Sunday) through 7 (Friday)
start_station_name	Name of the station where the rider started
start_station_id	Id of the station where the rider started
end_station_name	Name of the station where the rider ended his or her trip
end_station_id	Id of the station where the rider ended his or her trip
start_lat	Latitude coordinate of the starting place
start_lng	Longitude coordinate of the starting place
end_lat	Latitude coordinate of the ending place
end_lng	Longitude coordinate of the ending place
member_casual	Membership types including casual and annual

Per instruction of the project, the columns of ride_length and day_of_week have been inserted.

This is public data provided by Lyft Bikes and Scooters, LLC ("BikeShare), operating the City of Chicago's Divvy bicycle sharing service under the Data License Agreement: https://divvybikes.com/data-license-agreement.

Therefore, the data complies with ROCCC (Reliable, Original, Comprehensive, Current, and Cited).

Process

Following the instructions of the project, initially I opened each monthly CSV file in Excel fomat and added two columns : ride_length and day_of_weekcreated an Excel file from each csv file and inserted "ride_length" and "day_of_week" and tried to analyze the data in R using RStudio.

However, I found that opening and saving CSV files in Excel affected some date-time fields, causing inconsistencies in the dataset.

To ensure the data integrity, I went back to the original CSV files downloaded from the Divvy trip data (from January 2025 through December 2025) and recreated the dataset in R, and I generated the variables "ride_length, "month", and "day_of_week " in R before doing data cleaning and analysis.

I glanced at the dataset and found there were some missing values mainly in station-related fields including station names and station IDs. However, I did not find any missing values in the key variables for analysis including member_casual, started_at, ended_at, and ride_length.

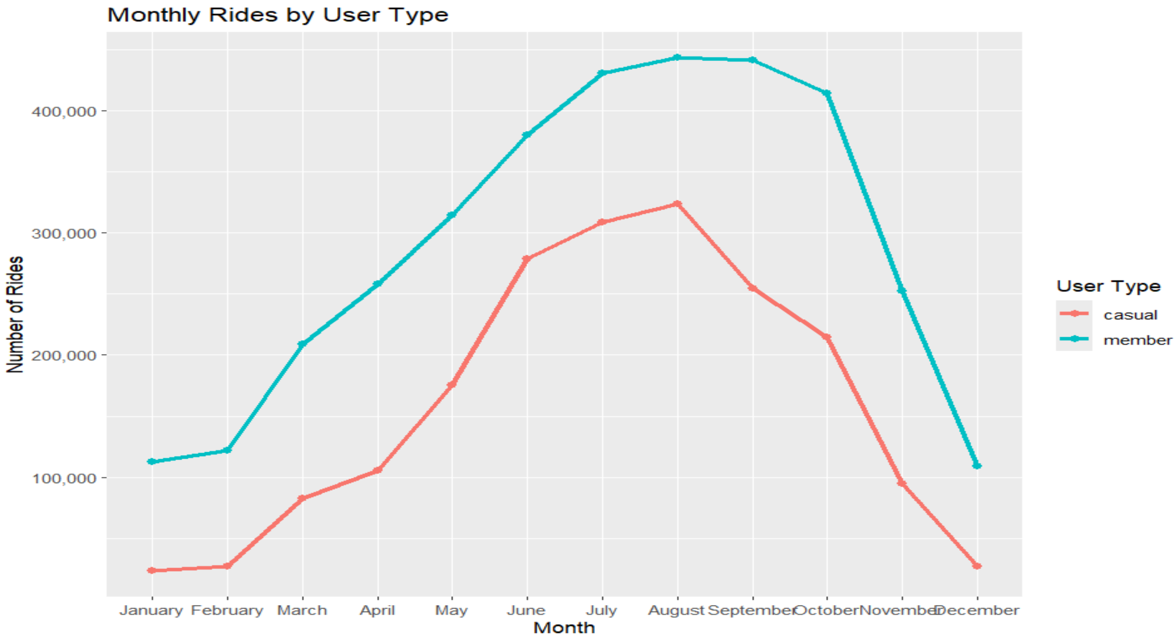
I used R and RStudio for as the analysis tool because the CSV files have so much data. For example, the January-CSV file has more than 100,000 lines of data. It is difficult to analyze the whole 12 month data in Excel. Using the R and RStudio, I found the original dataset contained 5,552,99 ride records. I removed the outlier, including rides shorter than one minute and longer than 24 hours, and that left 5,400,007 records for analysis. 152,987 records were excluded as potential outliers or invalid trips.

Analyze

I conducted a descriptive analysis of the dataset cleaned in the Process phase to address the business questions identified in the Ask phase. My focus was to determine whether the riding behavior was different between annual members and casual riders. The area of analysis includes ride frequency, average ride duration, weekly usage patterns, and seasonal trends.

The following are the four key findings in the differences between the two rider groups.

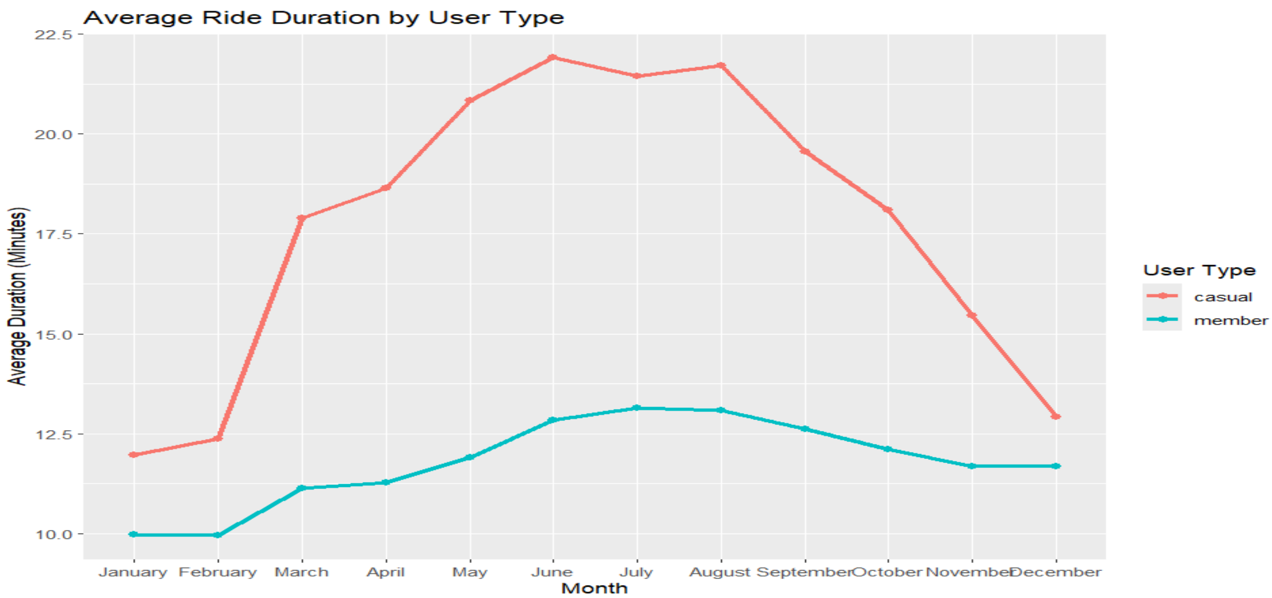
Figure 1. Monthly Rides by User Type



Key Insight 1:
Annual members took more rides than casual riders throughout the year.

Key Insight 2:
Both annual members and casual riders reached a peak in usage in August. However, the number of rides by casual riders dropped sharply after August.

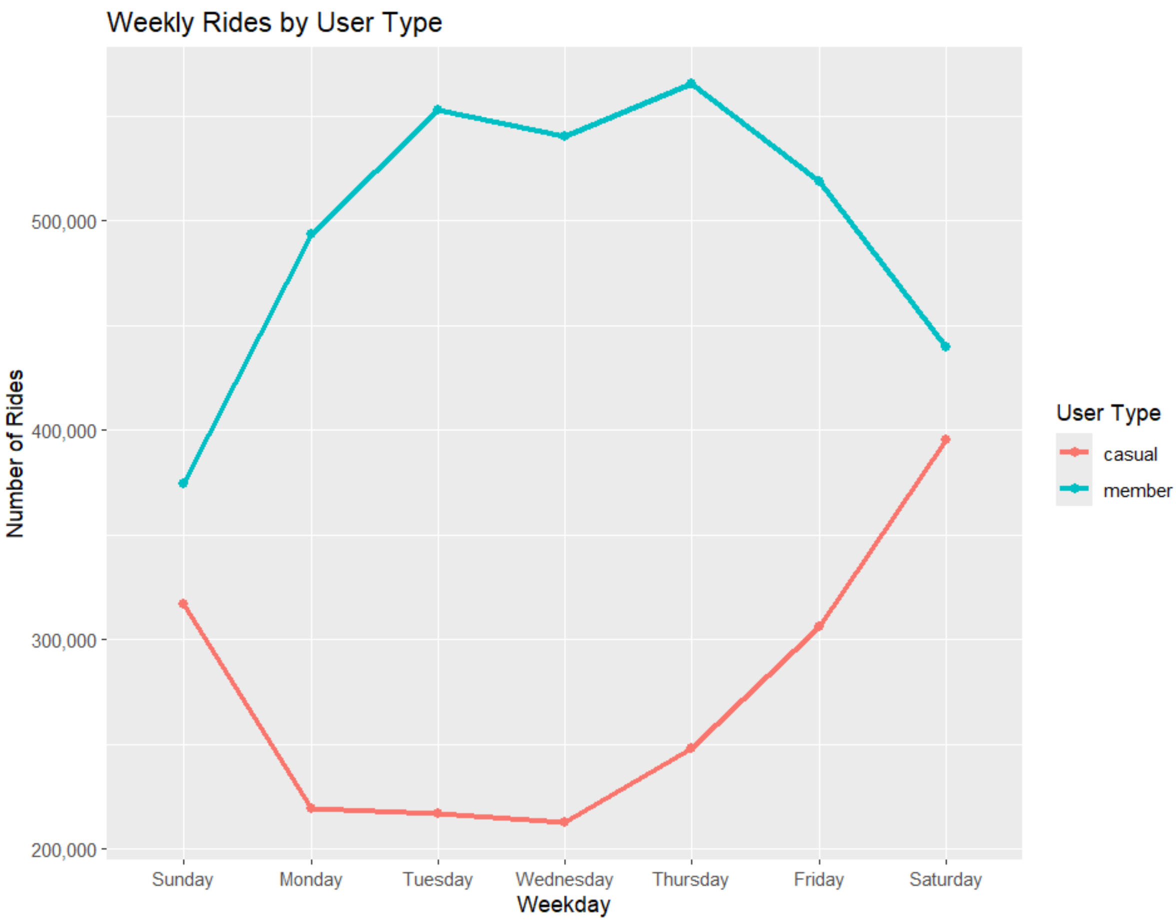
Figure 2. Average Ride Duration by User Type



Key Insight 3:

For each ride, casual riders spent more time than annual members.

Figure 3. Weekly Rides by User Type



Key Insight 4:
Annual members showed higher usage of bikes on weekdays, while casual riders used bikes more on weekends.

Act

Based on the analysis, I have two recommendations for converting casual riders into annual members.

Recommendation 1:
Promote annual memberships to frequent casual riders with incentives, including volume discounts. For example, Cyclistic could demonstrate how much more money frequent casual riders could save by purchasing an annual membership instead of paying for a ride each time.

Recommendation 2:
Promote annual memberships to casual riders during the summer season and weekends because their riding activities are at their peak.

References

Google Data Analytics Professional Certificate
Case study: How does a bike-share navigate speedy success?

Divvy Trip Data
https://divvy-tripdata.s3.amazonaws.com/index.html.

Divvy Bikes Data License Agreement
https://divvybikes.com/data-license-agreement